A Mini Project Report

on

Diabetic Retinopathy and Metabolic Factor: Explainable Deep Learning Screening with Multimodal Clinical Risk Assessment

Submitted in partial fulfilment of the requirements for the Major/Minor Data Science Program in the course AI in Medical Imaging Analysis

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DECLARATION

We hereby declare that the Mini Project Report entitled “Diabetic Retinopathy and Metabolic Factor: Explainable Deep Learning Screening with Multimodal Clinical Risk Assessment” which is being submitted to Symbiosis Artificial Intelligence Institute (SAII साई), Symbiosis International (Deemed University) in partial fulfillment of the requirements for the completion of Third semester in Bachelor of Business Administration (Artificial Intelligence) is a bonafide report of the work carried out by us. The material contained in the report has not been submitted to any University or Institution.

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CERTIFICATE

This is to certify that the Project Report entitled "Diabetic Retinopathy and Metabolic Factor: Explainable Deep Learning Screening with Multimodal Clinical Risk Assessment" submitted by Sanobar Shaikh (PRN: 25030421039), Srushti Lambdade (PRN: 25030421021) and Prajakta Suryawanshi (PRN: 25030421029) as record of the work carried out by them, is accepted as Report submission in partial fulfilment of the requirements for the completion of third semester in Bachelor of Business Administration (Artificial Intelligence).

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Abstract

Diabetic retinopathy (DR) is a complication of diabetes in which the small blood vessels of the retina are damaged. It often causes no symptoms in its early stages and is a leading cause of preventable vision loss among adults with diabetes. Screening programmes struggle to keep up because eye specialists are few and grading every fundus photograph by hand is slow. This project, DR Screening Suite (app name RetinaGuard AI), is a research prototype that supports early and explainable screening and also takes metabolic risk into account. A fundus photograph is resized and contrast-enhanced with CLAHE, and an EfficientNet-B0 classifier grades it from 0 (No DR) to 4 (Proliferative). Grad-CAM heatmaps show which regions influenced the grade. A U-Net draws lesion maps for microaneurysms, haemorrhages, hard exudates and soft exudates, and a ResNet locates the optic disc and the fovea. Separately, a rule-based score built from HbA1c, systolic blood pressure and diabetes duration gives a metabolic risk level. The two parts are combined at decision level, with 65% weight on the retinal grade and 35% on the metabolic level, and the result is presented in a Streamlit web application with a downloadable PDF report. On the 733 held-out photographs of the APTOS 2019 dataset (3,662 photographs in total) the classifier reached an accuracy of 84.6% and a quadratic weighted kappa (QWK) of 0.905. On the external IDRiD grading set of 516 photographs, accuracy fell to 78.8% and QWK to 0.847, which shows the effect of domain shift between datasets. Retinal images and clinical measurements come from different sources and are therefore combined by rules rather than learned jointly, and the lesion model was trained on only 54 images. The system is an educational research prototype and not a medical device.

Keywords: Diabetic Retinopathy, Deep Learning, EfficientNet-B0, CLAHE, Grad-CAM, Explainable AI (XAI), Multimodal Risk Assessment.

Chapter 1: Introduction

1.1 Background

Diabetic retinopathy is damage to the tiny blood vessels that supply the retina, the light-sensitive layer at the back of the eye. Persistently high blood sugar weakens these vessels, which may leak fluid and blood, close off, or be replaced by fragile new vessels. Because the central vision may remain normal for years, many patients are unaware of the disease until it is advanced, and by then some of the damage cannot be reversed.

The condition is assessed from a fundus photograph, a colour image of the retina taken through the pupil. Ophthalmologists look for microaneurysms, haemorrhages and exudates and assign a severity grade. In this project we use the five-level international scale [12]: grade 0 (No DR), grade 1 (Mild), grade 2 (Moderate), grade 3 (Severe) and grade 4 (Proliferative). The grade decides what happens next, whether that is a routine yearly check, a shorter follow-up interval or an urgent referral. An accurate and quick way of assigning the grade therefore has direct value for patient care.

1.2 Global Impact of Diabetic Retinopathy

Diabetic retinopathy affects a very large number of people. A pooled analysis of population studies estimated that about one in three people with diabetes show some signs of retinopathy [10], which is the figure that motivated our project. A later systematic review and meta-analysis puts the number of people living with the condition at over 100 million worldwide and projects that the burden will keep growing through 2045 as the number of people with diabetes rises [11].

The encouraging part is that most vision loss from DR can be prevented or delayed when the disease is found early and treated on time. This only works if screening reaches patients before symptoms appear, which is difficult in regions with few eye specialists. Automated screening tools that can look at a retinal photograph quickly are therefore seen as a practical way of extending the reach of existing clinics.

1.3 Motivation

Our motivation came from three observations. First, grading fundus photographs by hand is slow, and the number of trained graders is small compared with the number of people who need to be screened. A model that can pre-grade images allows specialists to concentrate on the cases that need attention.

Second, a bare grade is difficult to trust. A clinician wants to know why a model decided what it did, so we added visual explanations through Grad-CAM, as well as lesion maps and landmark markers that can be checked against the image.

Third, the state of the retina is only part of the picture. Blood sugar control, blood pressure and the number of years a person has lived with diabetes strongly influence how quickly the disease progresses. We therefore added a simple metabolic risk check so that the final recommendation reflects both the eye and the patient.

1.4 Objectives of the Study

The objectives of this mini project are:

1. To grade diabetic retinopathy from 0 to 4 from a fundus photograph using an EfficientNet-B0 classifier.
2. To explain each grade with Grad-CAM heatmaps that show the regions the classifier relied on.
3. To mark lesions with a U-Net and the optic disc and fovea with a ResNet.
4. To compute a metabolic risk level from HbA1c, systolic blood pressure and diabetes duration.
5. To combine both parts into one recommendation and a downloadable PDF report inside a web application.
6. To test the classifier on APTOS 2019 and on an external dataset, IDRiD, and to report the results honestly.

Chapter 2: Literature Survey and Medical Imaging Taxonomy

This chapter reviews the work our project builds on. Table 2.1 summarises the main approaches, what each does well, and where each falls short.

2.1 Deep learning for DR detection. The most influential early results came from large convolutional neural networks trained on hundreds of thousands of labelled photographs. Gulshan et al. showed that a deep network could detect referable retinopathy in fundus photographs with performance comparable to a panel of ophthalmologists [5]. Ting et al. extended this to a multi-ethnic population and to related eye diseases, which showed that such systems could work across different settings [6]. These studies confirm that grading can be automated, but they depend on very large labelled datasets that student projects do not have.

2.2 Transfer learning and modern backbones. When labelled medical data is limited, a common strategy is to start from a network pre-trained on a large natural-image collection and fine-tune it on the target task [14]. EfficientNet scales network depth, width and input resolution together with a fixed ratio and obtains good accuracy with relatively few parameters [1]. We chose its smallest version, EfficientNet-B0, because it is light enough to train and run

in Google Colab. Residual networks use skip connections that make deeper networks easier to train [4], and we use a ResNet backbone for the landmark model.

2.3 Vision Transformers. The Vision Transformer splits an image into patches and processes them with self-attention, which lets the model relate distant parts of the image [13]. Transformers usually need more data or heavier pre-training than convolutional networks, which makes them less convenient on a dataset of a few thousand retinal photographs, so we stayed with a convolutional classifier.

2.4 Segmentation, explanation and preprocessing. The U-Net combines an encoder that captures context with a decoder that restores resolution, joined by skip connections, and it is known to work with small annotated sets [3]. This is why we used it for lesion maps. Grad-CAM produces a coarse heatmap of the image regions that influenced a particular prediction without any retraining of the network [2]. Contrast-limited adaptive histogram equalisation (CLAHE) enhances local contrast while limiting noise amplification [7], and is widely used to make vessels and small lesions easier to see in fundus images. Severity is reported on the international five-grade scale [12], and agreement with reference grades is measured with the weighted kappa statistic [15].

2.5 Research gap. Many studies concentrate on a single model that outputs a grade. Few tools place retinal grading, a visual explanation, lesion maps and the patient's metabolic context in one application that a non-programmer can use. Our work attempts to fill this gap at the level of a student prototype by joining these parts in a single workflow, and by evaluating the classifier on a second dataset instead of only on its own validation split

Table 2.1: Approaches used in DR analysis (taxonomy)

Approach	Core idea	Strength	Limitation	Ref.
Deep CNN graders	Learn DR grades directly from many labelled fundus photos	Strong accuracy with large data	Needs large labelled datasets	[5], [6]
Transfer learning	Start from ImageNet-pre-trained weights and fine-tune	Works with smaller medical datasets	ImageNet features differ from retina images	[14]
EfficientNet family	Scale depth, width and resolution together	Good accuracy with few parameters	Still a black box on its own	[1]
Vision Transformers	Split the image into patches and use attention	Capture long-range context	Usually need more data or pre-training	[13]
U-Net segmentation	Encoder-decoder with skip connections	Works with small annotated sets	Tiny lesions are often missed	[3]
Grad-CAM	Gradient-weighted class activation map	Simple, needs no retraining	Coarse; shows attention, not lesions	[2]

Chapter 3: Dataset and Preprocessing

3.1 APTOS 2019 Dataset

The APTOS 2019 Blindness Detection dataset [9] contains 3,662 labelled fundus photographs, each graded from 0 to 4 by clinicians. We used it to train and validate the classifier, and 733 photographs were kept as a held-out validation set for evaluation.

The classes are strongly imbalanced. Table 3.1 and Figure 3.1 show that grade 0 accounts for 1,805 images, about half of the dataset, while Severe (193 images) and Proliferative (295 images) together make up only 13.4%. This imbalance influences both how the classifier is trained and how its results should be read.

Table 3.1: APTOS 2019 class distribution (percentages rounded)

Grade	Label	Photos	Share
0	No DR	1,805	49.3%
1	Mild	370	10.1%
2	Moderate	999	27.3%
3	Severe	193	5.3%
4	Proliferative	295	8.1%
	Total	3,662	100%

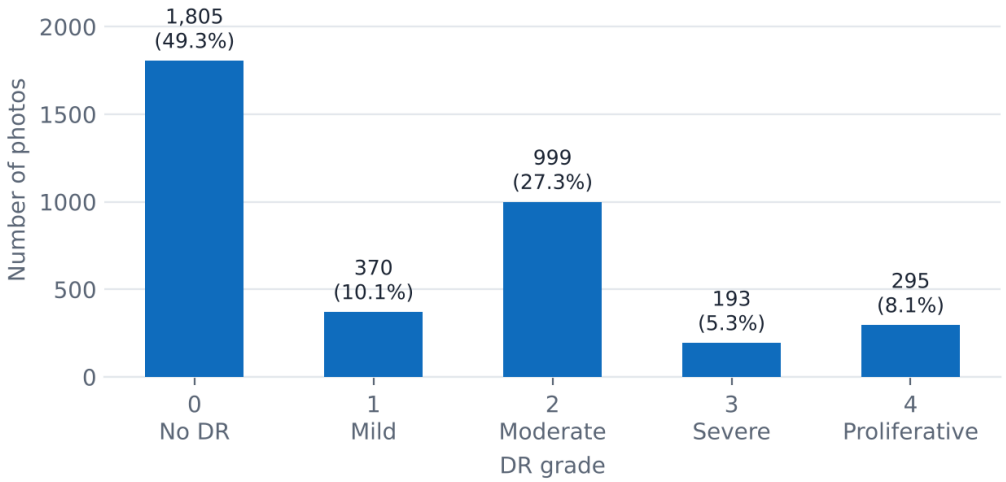


Figure 3.1: Class distribution of the 3,662 APTOS 2019 photos (counts and share of the dataset).

3.2 IDRiD Dataset

The Indian Diabetic Retinopathy Image Dataset (IDRiD) [8] was collected from an Indian population and has three parts: disease grading, lesion segmentation and optic disc and fovea annotation. Its photographs were taken with a different camera and from a different population than those in APTOS, which makes it a good external test: a model that has learnt only the quirks of APTOS will not perform well on it. Table 3.2 lists the parts we used and what each was used for. The 516 grading images were never used for training the classifier and serve purely as an external test set.

Table 3.2: IDRiD parts used in this project

Part	Photos	Use in our project
Disease grading (0-4)	516	External test set for the classifier
Lesion segmentation	81 (54 train / 27 test)	Training and testing the U-Net lesion maps
Optic disc and fovea points	516	Used for the ResNet landmark model

3.3 Class Imbalance

Table 3.1 shows that grades 3 and 4 are the rarest classes, with 193 and 295 images, against 1,805 for grade 0. A classifier trained naively on such data tends to favour the common classes, because predicting them most of the time already gives a decent accuracy. To limit this effect, rare grades were shown to the network more often during training. This does not create new information about the rare classes, so the weakest performance still appears on the Severe and Proliferative grades, as discussed in Chapter 6.

3.4 Image Preprocessing

All images pass through the same preprocessing steps, summarised in Table 3.3. Fundus photographs vary widely in brightness and contrast, and the signs of early disease, such as small red dots and faint vessel changes, can be hard to see. We therefore apply CLAHE, which equalises contrast in small tiles of the image and limits noise amplification [7]. This makes vessels and small lesions stand out more clearly for the networks.

Images are resized to 224 pixels for the grading network and to 512 pixels for the lesion network, since tiny lesions need more resolution. Pixel values are then normalised the way the pre-trained network expects. During training we added random flips, rotations and brightness changes, so that the model does not memorise individual images. The best epoch was selected on the held-out images using QWK. In the web application the contrast-enhanced and green-channel views can also be switched on for display only; these views are not fed to the models.

Table 3.3: Preprocessing and training steps

Step	What it does
Resize	224 px for grading; 512 px for lesion maps
CLAHE	Boosts contrast so vessels and small lesions stand out
Normalize	Scales pixel values the way the pre-trained network expects
Augmentation	Random flips, rotation and brightness changes so the model does not memorize
Imbalance handling	Rare grades are shown more often during training
Model selection	Train, check on held-out images, save the best epoch by QWK

Chapter 4: System Architecture and Methodology

4.1 Pipeline Overview

Figure 4.1 shows the overall pipeline. The system has two branches that are scored independently and joined only at the end. In the retinal branch, an uploaded fundus photograph is resized and enhanced with CLAHE. It is then passed to three models: EfficientNet-B0 predicts the DR grade (0 to 4) and Grad-CAM produces a heatmap for that prediction, the U-Net produces lesion maps, and the ResNet predicts the positions of the optic disc and the fovea.

In the metabolic branch, the user enters HbA1c, systolic blood pressure and years with diabetes. A rule-based score converts these values into a metabolic risk level. Finally, decision-level fusion combines the retinal grade and the metabolic level into one combined risk, which is shown in the application and written into the PDF report. Table 4.1 lists the three neural networks, their tasks and the data they were trained on.

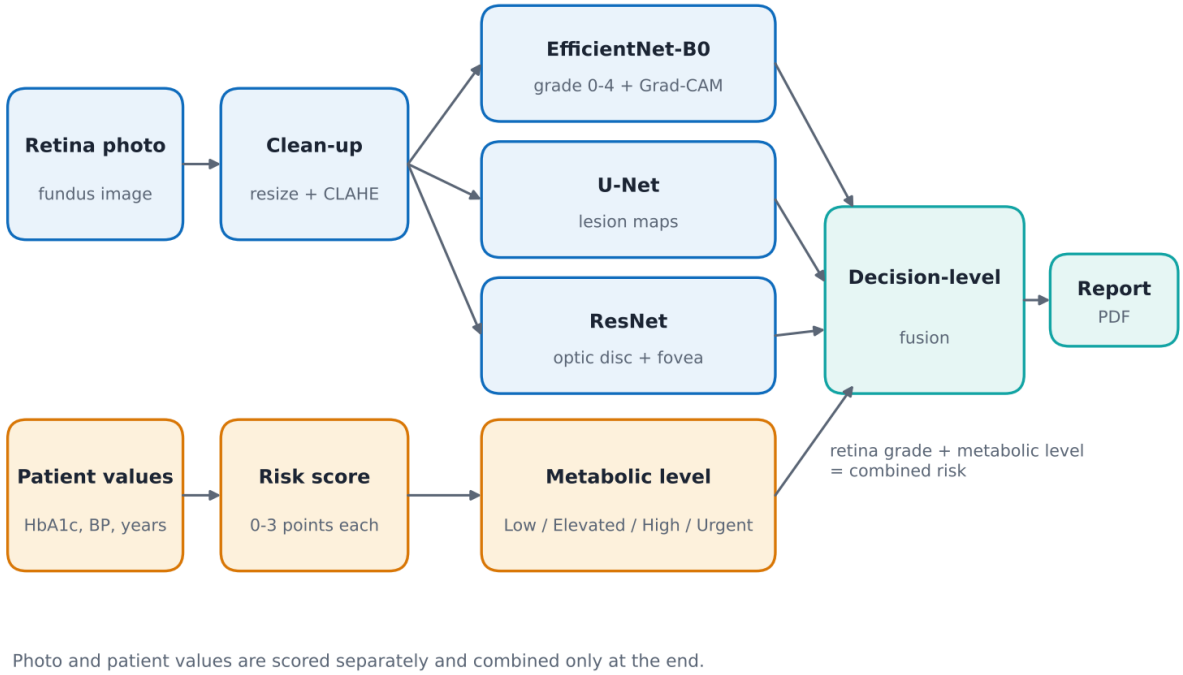


Figure 4.1: Pipeline of DR Screening Suite. The retina branch (blue) and the metabolic branch (amber) are scored separately and combined by decision-level fusion (teal).

Table 4.1: Models in the pipeline

Model	Task	Trained on	Output in the app
EfficientNet-B0	Grade DR from 0 to 4	APTOS 2019	Grade, probabilities, Grad-CAM heatmap
U-Net	Segment lesions	IDRiD lesion set (54 train images)	Lesion maps you can switch on and off
ResNet	Locate landmarks	IDRiD optic disc and fovea points	Optic disc and fovea markers

4.2 EfficientNet-B0 Classifier

The classifier is an EfficientNet-B0 network [1] that was pre-trained on ImageNet and fine-tuned on APTOS 2019 for five output classes. Transfer learning is useful here because the pre-trained layers already detect edges, textures and shapes, so training can concentrate on retinal patterns [14] instead of starting from random weights.

Four ideas shaped the training. We started from pre-trained weights, we corrected for class imbalance by showing the rare grades more often, we used augmentation to add variety, and we kept the epoch with the best QWK on the held-out images. The input size is 224 by 224 pixels.

We select models by QWK because DR grades are ordered. QWK measures the agreement between the predicted and the reference grades and penalises disagreements by the square of their distance [15], so predicting grade 0 for a patient with grade 4 costs far more than

predicting grade 1 for a patient with grade 2. A value of 1 means perfect agreement. The classifier outputs a probability for each grade, and the grade with the highest probability is the prediction.

4.3 U-Net Lesion Segmentation

Lesion maps are produced by a U-Net [3], an encoder-decoder network whose skip connections carry fine detail from the encoder to the decoder. This design is well suited to small annotated datasets. We trained it on the IDRiD lesion-segmentation part, which has 81 images, of which 54 were used for training and 27 for testing.

The network marks four lesion types: microaneurysms, haemorrhages, hard exudates and soft exudates. Each type can be switched on and off in the application, and the opacity of the overlay can be changed. Because the training set is very small and microaneurysms cover only a few pixels, these lesions are often missed, so the maps should be treated as an aid for the viewer and not as a precise measurement.

4.4 ResNet Landmark Localization

The optic disc and the fovea are the main anatomical landmarks of the retina. The optic disc is where the nerve fibres leave the eye, and the fovea is the centre of sharp vision. Lesions close to the fovea threaten sight the most, which is why their positions are useful. We trained a ResNet regression model [4] on the IDRiD optic disc and fovea annotations, which cover 516 images. The network predicts the coordinates of both landmarks, and the application draws them as labelled circles on the photograph.

4.5 Metabolic Risk Score

The metabolic score is a simple rule-based calculation and not a trained model. It uses three inputs: HbA1c, which reflects average blood sugar over the previous few months, systolic blood pressure, and the number of years the person has lived with diabetes. Each input is converted to 0 to 3 points according to increasing risk bands, so the total score lies between 0 and 9. Table 4.2 shows how the total is mapped to four levels: Low (0 to 2), Elevated (3 to 4), High (5 to 6) and Urgent (7 to 9).

We used rules because no public dataset provides retinal images together with blood measurements for the same patients, so a learned model was not possible. The rule-based approach is transparent, since a user can see exactly how each value contributes, but the cut-offs are a simplification and not a validated clinical risk model.

4.6 Decision-Level Fusion

The retinal grade and the metabolic level are not trained together. Public retina datasets do not contain linked blood tests, so the two parts are scored separately and are combined only at the end, which is what is meant by decision-level fusion.

The combined risk is a weighted combination of the two results, with 65% weight on the retinal grade and 35% on the metabolic level. The retinal grade carries more weight because it is a direct observation of the disease, whereas the metabolic score describes the surrounding risk. Two safety rules override the weighting, as shown in Table 4.2: a Severe grade (3) always leads to a combined risk of at least High, and a Proliferative grade (4) always leads to Urgent. These rules make sure that a bad retinal finding cannot be diluted by good blood values. The final level is shown in the Combined Report tab with a short advice text and is included in the PDF.

Table 4.2: Metabolic levels and fusion rules[12]

Metabolic level	Total score (0-9)
Low	0-2
Elevated	3-4
High	5-6
Urgent	7-9

Fusion rule	Value
Weighting	65% retina grade + 35% metabolic level
Severe (grade 3)	Combined risk is at least High
Proliferative (grade 4)	Combined risk is always Urgent

Chapter 5: Explainable AI and Clinical Decision Support

5.1 Grad-CAM Visual Explanation

Grad-CAM [2] explains a prediction by showing which parts of the image the network used to reach it. In our application it is computed for the grade that the classifier predicts. First, the image is passed through the network. Next, the gradient of the predicted grade is calculated with respect to the feature maps of the last convolutional layer. These gradients are averaged over each feature map to give one weight per channel. The feature maps are then combined as a weighted sum, and a ReLU keeps only the regions that push the score towards the predicted grade. Finally the result is normalised to the range 0 to 1, resized to the size of the photograph and overlaid on it as a colour heatmap.

No retraining is needed, because the method only uses the gradients that the network already provides. The resulting heatmap is coarse, since the last convolutional layer has a low spatial

resolution, so it indicates regions and not exact lesion boundaries. The Explainability tab shows the original photograph and the heatmap side by side, with an opacity slider and a choice of colour maps.

5.2 Reading Warm and Cool Colours

In the heatmap, warm colours such as red and yellow mark the areas that pushed the model most strongly towards its predicted grade, while cool colours such as blue mark areas with little influence. A user should therefore look at whether the warm regions fall on the retina and on plausible findings, and not on the black border or an image artefact. The opacity slider in the application blends the heatmap with the photograph, so that the underlying anatomy remains visible.

5.3 Lesion Maps and Landmarks

Lesion maps and landmarks are separate outputs, and they do not come from Grad-CAM. The U-Net colours the four lesion types, microaneurysms, haemorrhages, hard exudates and soft exudates, and each layer has its own checkbox so that the viewer can look at one type at a time. The ResNet marks the optic disc and the fovea with labelled circles. It is important to keep these apart: Grad-CAM tells us where the grader looked, while the U-Net and ResNet outputs tell us where lesions and landmarks are predicted to be. When a model fails to load, the application falls back to a simple image-processing preview and labels it clearly as a preview, so that it is never mistaken for a trained-model output.

5.4 Clinical Justification and Limits

Explanations are useful for clinical decision support because they let a clinician check the model instead of simply trusting it. If a Severe grade is accompanied by a heatmap and lesion maps that point at real haemorrhages or exudates, the finding becomes easier to accept. If the heatmap highlights an irrelevant area, that is a sign that the grade should be treated with caution.

The limits must also be understood. A heatmap shows what the model attended to; it is not a diagnosis, and it can highlight areas that are not lesions. The whole system is a research prototype for screening assistance and does not replace an examination by an ophthalmologist.

Chapter 6: Results, Performance Metrics and Discussion

6.1 Evaluation Metrics

We report two main measures. Accuracy is the share of photographs for which the predicted grade equals the reference grade. It is easy to read, but with imbalanced classes it can look good even when rare grades are handled badly. Quadratic weighted kappa (QWK) [15]

measures the agreement between predicted and reference grades while penalising large errors more than small ones, with 1.0 meaning perfect agreement. Since DR grades are ordered, QWK is the more informative single number for this task.

A confusion matrix tabulates true grades in rows and predicted grades in columns, so that every kind of error can be seen. From it, a one-versus-rest view can be derived for each grade: sensitivity is TP / (TP + FN), the share of images of that grade that were found, and specificity is TN / (TN + FP), the share of images of other grades that were correctly left out. The Model Performance tab of the application shows the confusion matrix for the APTOS validation set (Figure 7.1).

6.2 Results on APTOS and IDRiD

Table 6.1 and Figure 6.1 show the results side by side. On the held-out APTOS 2019 validation set of 733 photographs the classifier achieved an accuracy of 84.6% and a QWK of 0.905, so it agrees closely with the reference grades, and most errors are between neighbouring grades. On the external IDRiD grading set of 516 photographs, which the model never saw during training, accuracy dropped to 78.8% and QWK to 0.847.

The two results answer different questions. The APTOS figure says how well the model works on data like its training data, and the IDRiD figure says how well it carries over to images from a different source. The large gap between them is the main finding of our evaluation and is discussed in Section 6.5.

Table 6.1: Classifier results

Dataset	Role	Photos	Accuracy	QWK
APTOS 2019	Held-out validation	733	84.6%	0.905
IDRiD	External test	516	78.8%	0.847

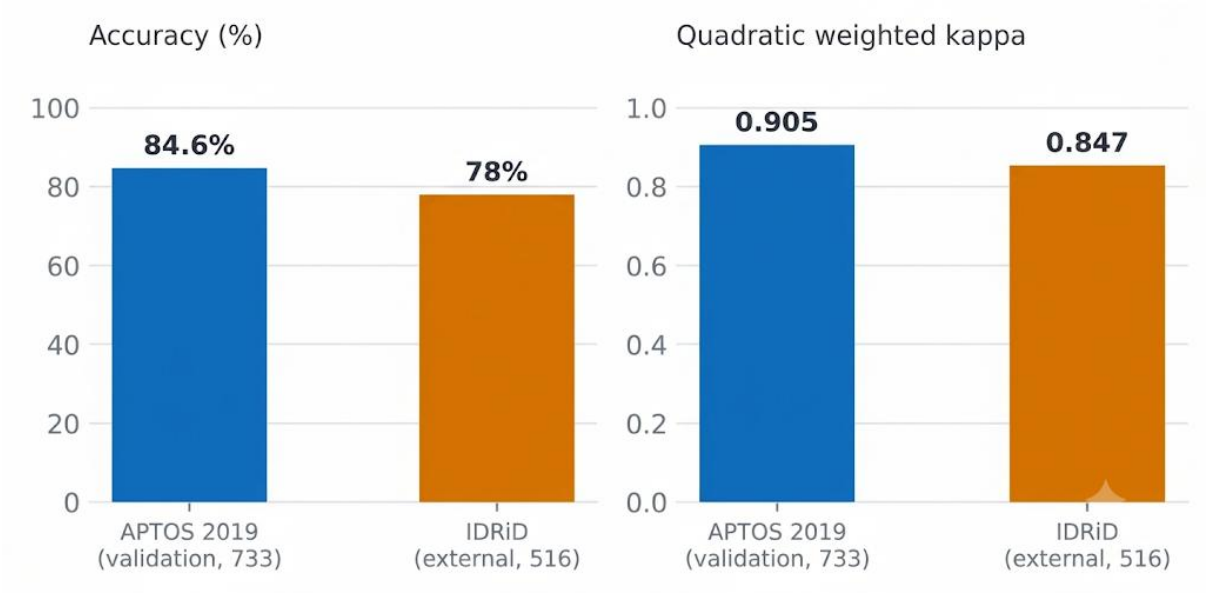


Figure 6.1: Accuracy and quadratic weighted kappa on the APTOS 2019 held-out split and on the external IDRiD set.

6.3 Per-Grade Analysis

On APTOS, the grades with the fewest training examples, Severe and Proliferative, are the weakest. This is expected from the class distribution in Table 3.1: the classifier sees far fewer examples of grades 3 and 4 and therefore learns them less reliably, even though the rare grades were shown more often during training. The confusion matrix in the Model Performance tab (Figure 7.1) also shows that most errors fall between neighbouring grades, such as No DR and Mild or Mild and Moderate, which are visually close and are also hard for human graders.

This pattern matters for screening, since the rare grades are the ones that most urgently need referral. In the combined risk, the safety rules for grades 3 and 4 (Section 4.6) partly address this, because a Severe or Proliferative prediction forces a high or urgent combined risk, but the classifier itself still needs more examples of these grades. Complete per-grade sensitivity and specificity values should be computed from the full confusion matrix in the saved evaluation output using the one-versus-rest definitions given in Section 6.1.

6.4 Training Behaviour

The Model Performance tab shows the training and validation curves recorded during training (Figure 7.1). The training loss falls steadily over the epochs, and the validation accuracy and QWK rise and then level off, which indicates that the model learns the task and stops improving after a number of epochs. The checkpoint with the best validation QWK was kept for the application.

6.5 Discussion: Domain Shift

The drop from 84.6% accuracy and 0.905 QWK on APTOS to 78.8% accuracy and 0.847 QWK on IDRiD is a domain shift, not a coding error. The classifier was trained only on APTOS, and IDRiD differs in several ways: a different camera and imaging set-up, different lighting and colour balance, a different patient population, and possibly different grading practice and a different mix of grades. A network that has learned the appearance of APTOS images may rely on cues that do not transfer.

The fact that QWK stays at 0.847, far above zero, shows that the model still captures some real ordering of severity on the new data, even though its exact grades are often wrong. Accuracy falls much faster than QWK because many errors are between neighbouring grades, which accuracy counts as fully wrong but QWK penalises lightly.

The lesson is practical: a model that performs well on its own validation split should not be assumed to work on images from another clinic. Before real-world use it needs validation, and probably fine-tuning, on local data. We kept this finding visible in the application's Model Performance tab so that users see it too.

6.6 Limitations

The following limitations apply to our work. Retinal photographs and blood values come from different data sources, so they are combined by rules and not learned together, and the 65/35 weighting and the score cut-offs were chosen by design, not fitted to outcomes. The lesion model was trained on only 54 images, and tiny microaneurysms are often missed. The classifier performs much worse on the external IDRiD set than on APTOS, so its results cannot be assumed to hold for other cameras or populations. For the lesion and landmark models we have not computed measures such as Dice score or pixel error, so their accuracy is described only qualitatively. Finally, the system is a research prototype for learning. It is not a medical device, it has not been clinically validated, and it must not be used to make diagnostic decisions.

Chapter 7: Web Application and System Implementation

7.1 Tools and Technologies

The prototype is written in Python. PyTorch and torchvision provide the neural networks, OpenCV handles image processing and overlays, and Streamlit provides the interface. ReportLab generates the PDF report. Training and running the application took place in Google Colab, a shared Google Drive folder kept the files of the three team members together, and a Cloudflare tunnel gave the running application a public link. Table 7.1 lists the tools and their roles.

Table 7.1: Tools used

Tool	Use in the project
Python and PyTorch	Building, training and running the models
Streamlit	Interactive web app
OpenCV	Image processing and overlays
ReportLab	PDF report generation
Google Colab	Training and running the app
Google Drive	Shared project folder for all three members
Cloudflare tunnel	Public link to the running app

7.2 App Features

The application, called RetinaGuard AI, has five tabs. Retina Analysis shows the original image next to the image with lesion and landmark overlays, together with the predicted grade, the advice for that grade and the probability of each grade. Explainability shows the Grad-CAM heatmap with an adjustable opacity. Metabolic Risk takes HbA1c, systolic blood pressure and diabetes duration through sliders and shows the score out of 9 and the risk level. Combined Report shows the retinal grade, the metabolic level and the combined risk, and lets the user download the PDF. Model Performance presents the evaluation results, the domain-shift finding, the confusion matrix and the training curves; this tab is shown in Figure 7.1.

A sidebar lets the user upload a photograph or choose a sample, switch between the original, contrast-enhanced and green-channel views, change the heatmap opacity, colour map and lesion-mask opacity, toggle the optic disc and fovea markers, and choose a colour theme. It also lists whether each of the three models has loaded. The interface is responsive and can be used on a phone as well as on a computer.

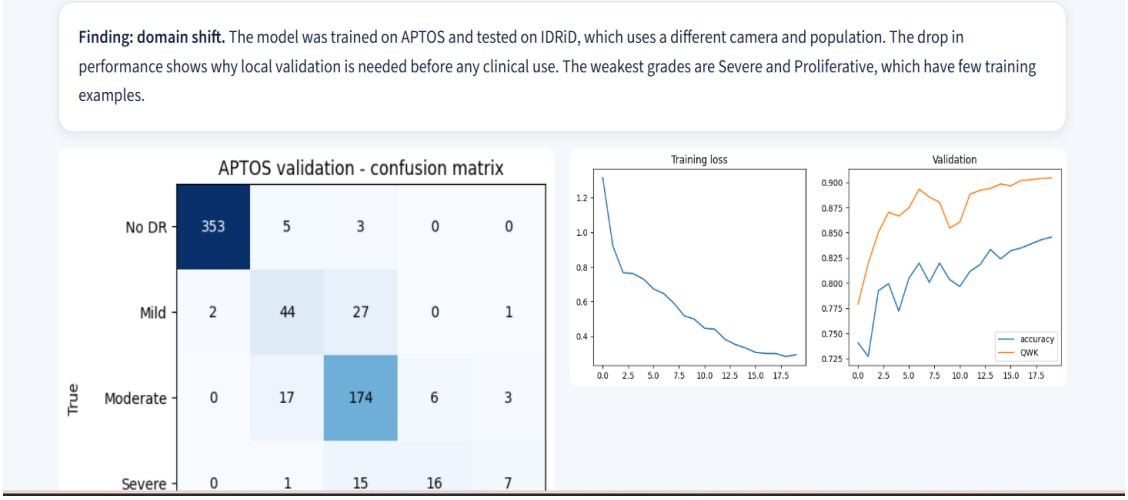
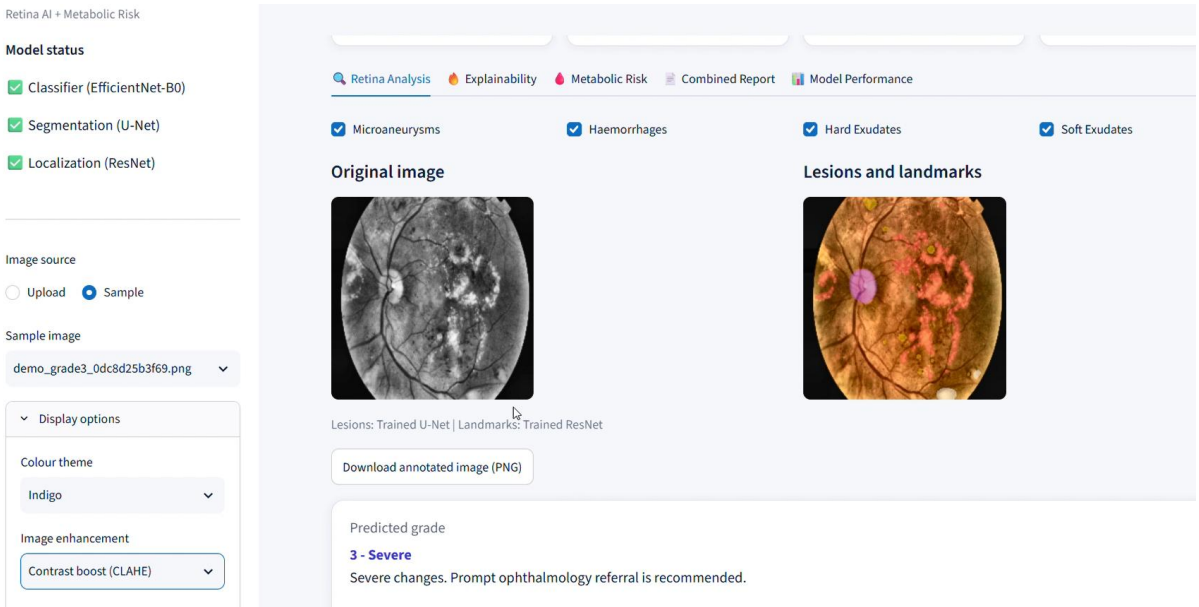


Figure 7.1: Model Performance tab of the web application, showing the evaluation metrics, the domain-shift finding, the confusion matrix (partly visible) and the training curves.

7.3 User Workflow

A typical session follows these steps:

1. Open the application link and upload a fundus photograph, or pick one of the sample images.
2. Read the predicted grade, the advice and the probabilities in the Retina Analysis tab, and switch lesion layers and markers on or off.
3. Open the Explainability tab to see which regions influenced the grade.
4. Enter HbA1c, systolic blood pressure and diabetes duration in the Metabolic Risk tab and press Calculate.
5. Open the Combined Report tab to see the combined risk and download the PDF report.



7.4 Report Generation and Deployment

The PDF report is generated with ReportLab. It contains the patient details, the predicted retinal grade with its probabilities, the metabolic inputs and level, the combined risk with its advice, the Grad-CAM image and a note stating that the report is a research output and not a diagnosis.

The application runs inside a Google Colab session and is exposed through a public Cloudflare tunnel link. This is convenient for demonstrations, but the link changes whenever the Colab session restarts, so it is not suitable for permanent deployment. The version described in this report is RetinaGuard AI, the premium Streamlit interface, which loads all three trained models and shows a clear status when one of them is unavailable.

Chapter 8: Future Scope and Conclusion

8.1 Future Scope

Several improvements are planned but have not been carried out. The most important is to close the gap between APTOS and IDRiD by fine-tuning on mixed APTOS and IDRiD data with stronger augmentation, and then to re-evaluate on a held-out IDRiD split. The rare Severe and Proliferative grades need more targeted training, for example with additional data or class-aware sampling and loss functions.

The lesion and landmark models should be evaluated properly, using the Dice score for lesion maps and pixel error for landmarks, and the lesion model would benefit from more annotated images. The metabolic score could be checked against clinical guidance with the help of a doctor.

On the deployment side, the application should be hosted permanently and tested with real users, and a lightweight version could run on low-power devices at screening camps. In the longer term, a prospective clinical study with linked retinal images and clinical measurements would be needed before any claim about real-world benefit can be made.

8.2 Conclusion

We built DR Screening Suite, a research prototype that grades diabetic retinopathy from fundus photographs, explains the grade with Grad-CAM heatmaps, marks lesions and landmarks with a U-Net and a ResNet, and combines the retinal result with a rule-based metabolic risk level into one recommendation and a PDF report, all inside a Streamlit web application.

The classifier achieved an accuracy of 84.6% and a QWK of 0.905 on the 733 held-out APTOS 2019 photographs. On the external IDRiD set of 516 photographs, accuracy was 78.8% and QWK 0.847. We regard this second result as the most honest and valuable finding of the

project, because it demonstrates that strong internal results do not guarantee that a model transfers to a new camera and population.

The main limitations are the weaker performance on rare grades, the small lesion training set, and the fact that retinal images and metabolic data were combined by rules and not learned jointly. The next steps are fine-tuning on mixed data, measuring the lesion and landmark models, and local validation. The project also taught the team how a complete pipeline, from data and models to explanation and a usable application, fits together.

8.3 Team Contributions

Table 8.1: Who did what

Member	Role	Contribution
Srushti Lambdade	Data and vision models	Data pipeline (APTOS + IDRiD), preprocessing (CLAHE), lesion and landmark models
Prajakta Suryawanshi	Classifier	Trained EfficientNet-B0, tested on APTOS and IDRiD, Grad-CAM and evaluation plots
Sanobar Shaikh	App and risk	Streamlit app, risk score and fusion rules, PDF report and deployment

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